SurfacePrint

Typographic Fingerprints of LLMs



Yuntian Deng

Same word. Different apostrophe.

don't

Straight

don’t

Curly

Count the formatting.



3 / 5

60% curly

A random forest learns formatting patterns.

Raw text → 97 features → Random forest → Score

Test new text from familiar sources.

TRAIN

Human text

+ **17 AI models**

TEST

New texts

from the same sources

Does generated text rank higher?

Human text < Generated text

Detector score

0.5

Chance

AUROC

1.0

Perfect

Formatting separates these benchmark texts.

0.983

Mean AUROC

Held-out texts · 17 known model sources

Some signal survives whitespace cleanup.

Trained on raw text

```
First paragraph.


Second paragraph.
```

0.983 AUROC

Trained on cleaned text

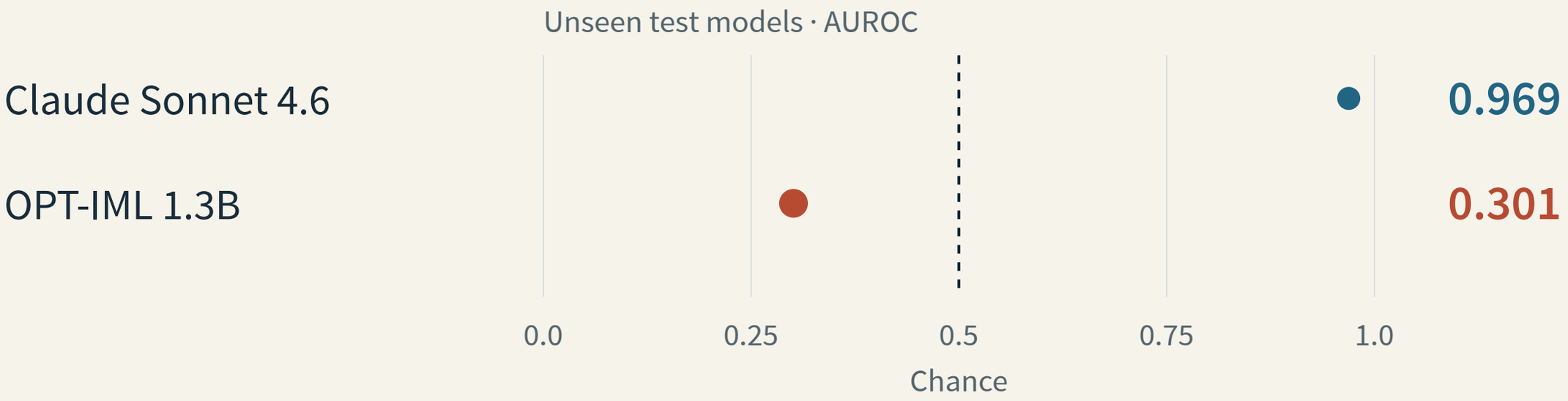
```
First paragraph.

Second paragraph.
```

0.889 AUROC

Transfer to new models is uneven.

Train: human + Claude Sonnet 4 · cleaned text



Same wording. Different formatting.

Before

```
**A simple answer**

It doesn’t change.
```

After

```
A simple answer

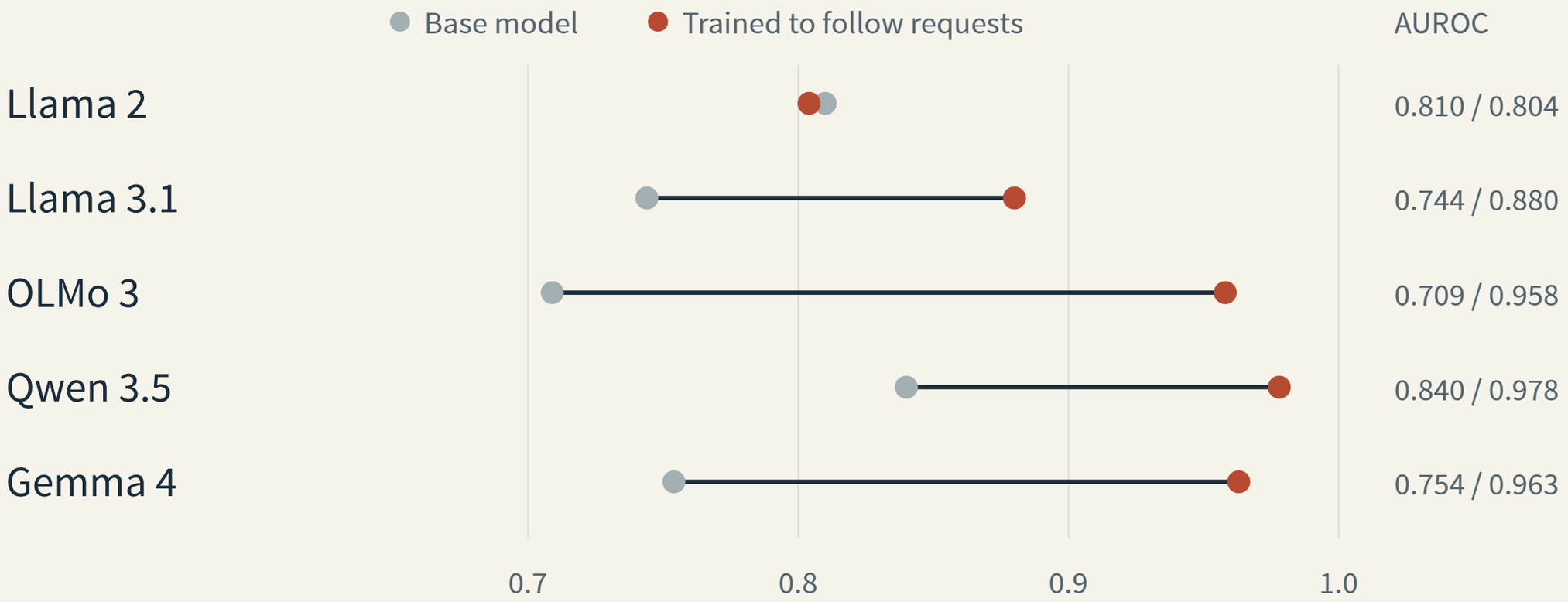
It doesn't change.
```

How often would
human text
be wrongly flagged?

Formatting is an
additional detection clue.

Instruction-tuned text is often easier to detect

Separate detectors · cleaned text · prompt templates also differ



An AUROC score does not set the false-positive rate

A cutoff turns a score into a flag

Different cutoffs trade missed AI text against wrongly flagged human text

The classifier combines many small decision trees

300	6	3
trees	maximum depth	minimum leaf size

97 measurements, balanced class weights, fixed seed

One path in an illustrative tree

Depth-3 tree · AUROC 0.750

1	Markdown bold present? **Yes**
2	Double-backtick inline code present? **No**

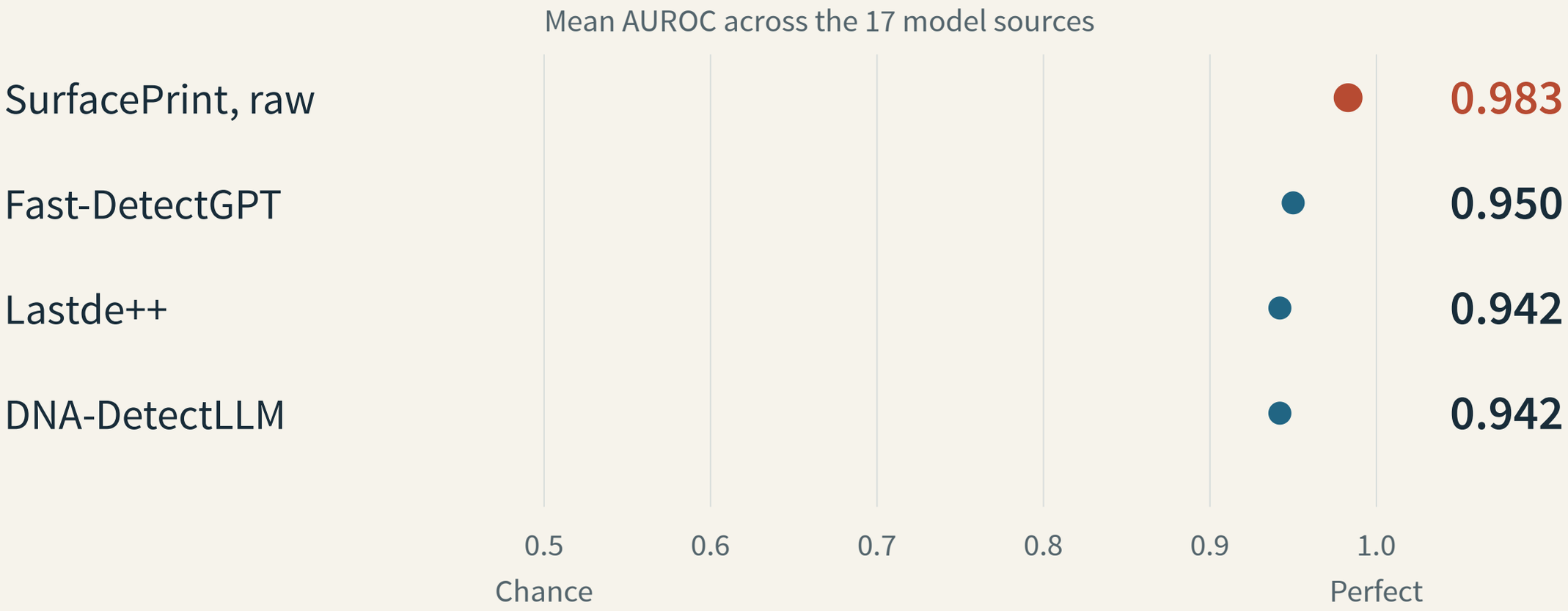
Result Tree predicts **LLM**

The 97 measurements cover five surface families

FAMILY	EXAMPLE	COUNT
Prose typography	' ’ ... …	22
Markdown	**bold** __bold__	25
Math markup	x \(x\)	24
Code formatting	Tabs spaces	20
Whitespace	Blank lines	6

The main reference-model comparators

SurfacePrint: supervised · Comparators: training-free



The source pool and detector sample differ

Source pool, per domain

2,000 train / 500 validation / 1,000 test prompts

Main detector’s training sample

8,000 human posts + 470 outputs per AI model

17 × 470 = 7,990 generated examples

SP-WS standardizes selected whitespace conventions

Line breaks	Use one line-break encoding
Spaces at line ends	Remove them
Repeated blank lines	Keep one blank line
Final line break	Remove it

Punctuation and markup remain in the input

Proposed: test real document handling

Representative documents

Human and generated text from the intended setting

Real processing steps

Include ordinary editing, copying and extraction

Held-out error measurement

Report false positives and detection at a fixed cutoff